Magnetospheric Extension: Curvature Variable Physics Applied to the Van Allen Radiation Belts

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206 Innovation Inc. • Bellevue, Washington, USA • April 2026

Abstract— This manuscript develops a standalone magnetospheric extension of Curvature Variable Physics (CVP) for application to the Van Allen radiation belts. The central proposition is that classical radiation-belt models, while successful in many forecasting and interpretive settings, treat the magnetosphere using a sufficiently smooth geometric baseline that dominant local curvature gradients of storm-time transport, wave-particle scattering, and trapping-manifold distortion are not represented as first-class state variables. In the proposed extension, magnetic curvature is promoted to a dynamic field K_B(t, L, alpha, phi), where L denotes McIlwain L-shell, alpha pitch angle, and phi magnetic longitude. A magnetospheric divergence operator D_B is then defined as the instability associated with omitted curvature gradients, and a magnetic response operator C_B(K_B) = beta1 dK_B/dt + beta2 grad K_B + beta3 Laplacian K_B is introduced as the formal corrective layer. The paper states the mathematical grammar of the extension, aligns the operator structure with known radiation-belt processes such as radial diffusion and chorus-driven scattering, and outlines a GPU-accelerated implementation path for storm-time simulations. The manuscript does not claim empirical validation of CVP-B; rather, it presents a coherent formalism and computational research program for evaluating whether curvature-variable closure can regularize magnetospheric transport in high-consequence space-weather environments.

Keywords— Curvature Variable Physics; Van Allen radiation belts; magnetosphere; radiation-belt transport; space weather; curvature-response operator; Omega-Genesis.

Nomenclature

Symbol	Meaning
K_B(t, L, alpha, phi)	Magnetospheric curvature field over time, L-shell, pitch angle, and longitude
D_B	Magnetic divergence operator representing omitted-gradient instability
C_B(K_B)	Magnetic curvature-response operator
Phi_B	Magnetospheric syntropy potential
L	McIlwain shell parameter
alpha	Pitch angle of trapped particles

1. Introduction

The Van Allen radiation belts are persistent toroidal populations of energetic charged particles trapped by Earth's geomagnetic field. Their variability matters directly for spacecraft charging, mission design, astronaut safety, and the forecasting of hazardous space-weather conditions. NASA's Van Allen Probes mission produced nearly seven years of in situ measurements and documented a highly structured and rapidly evolving environment, including transient third-belt behavior and rapid reconfiguration of outer-belt electron populations during geomagnetic activity.

Modern radiation-belt theory already includes sophisticated diffusion, transport, and wave-particle interaction mechanisms. Nevertheless, the governing models are usually expressed in terms of smooth magnetic geometry, diffusion coefficients, source and loss operators, and boundary conditions. The present manuscript proposes a different interpretive layer: magnetospheric stability may be usefully reformulated through a curvature-variable field model in which the geometry of the trapping manifold becomes a dynamic state variable rather than a passive background.

Curvature Variable Physics (CVP) was introduced in prior gravitational work as a response to static-curvature omission in irregular-body descent. This paper extends that operator grammar to the magnetosphere. The objective is not to replace the established radiation-belt literature, but to state a formal extension that can be reviewed, tested computationally, and compared against measured storm-time flux evolution.

2. Background: Radiation-Belt Dynamics and Modeling Context

Earth's radiation belts are shaped by magnetic trapping, radial transport, local acceleration, pitch-angle scattering, and loss to the atmosphere and magnetopause. The outer belt is especially dynamic: relativistic electron fluxes can intensify or deplete by orders of magnitude during geomagnetic storms, with chorus waves, hiss, radial diffusion, and magnetopause shadowing all contributing to the observed variability.

The Van Allen Probes and related missions such as THEMIS established that outer-belt response is highly structured in space and time. Pitch-angle distributions evolve nontrivially, chorus wave morphology varies with local time and geomagnetic conditions, and flux changes can be rapid enough to challenge simple static interpretations of the trapping environment.

The classical reduced transport description is often written as a Fokker-Planck-type equation for phase-space density f, in which diffusion, sources, and losses govern time evolution. This manuscript keeps that heritage in view but introduces a curvature-variable overlay: instead of treating magnetic geometry as an implicitly smooth support for transport, it treats the geometry of the trapping manifold as an explicit dynamic field subject to gradients, curvature transitions, and response operators.

3. Field Definition: Magnetospheric Curvature as a Dynamic Variable

The proposed extension begins by defining a magnetospheric curvature field $K_B = K_B(t, L, \alpha, \phi)$, where t denotes geomagnetic time, L the McIlwain shell parameter, alpha the local pitch angle, and phi the magnetic longitude. In this formulation, K_B does not mean curvature in the narrow differential-geometric sense alone. It denotes an effective curvature field of the trapping manifold, constructed from the geometry and gradients that govern particle stability. Operationally, such a field may be assembled from magnetic-field structure, plasma-density gradients, wave intensity, drift-shell splitting, and storm-time compression.

This definition plays the same formal role that $K(t, r, \theta, \phi)$ plays in the gravitational CVP formulation. The magnetosphere is therefore treated as a curvature-variable electromagnetic manifold whose stability depends not only on the presence of gradients but on how those gradients evolve and are regularized.

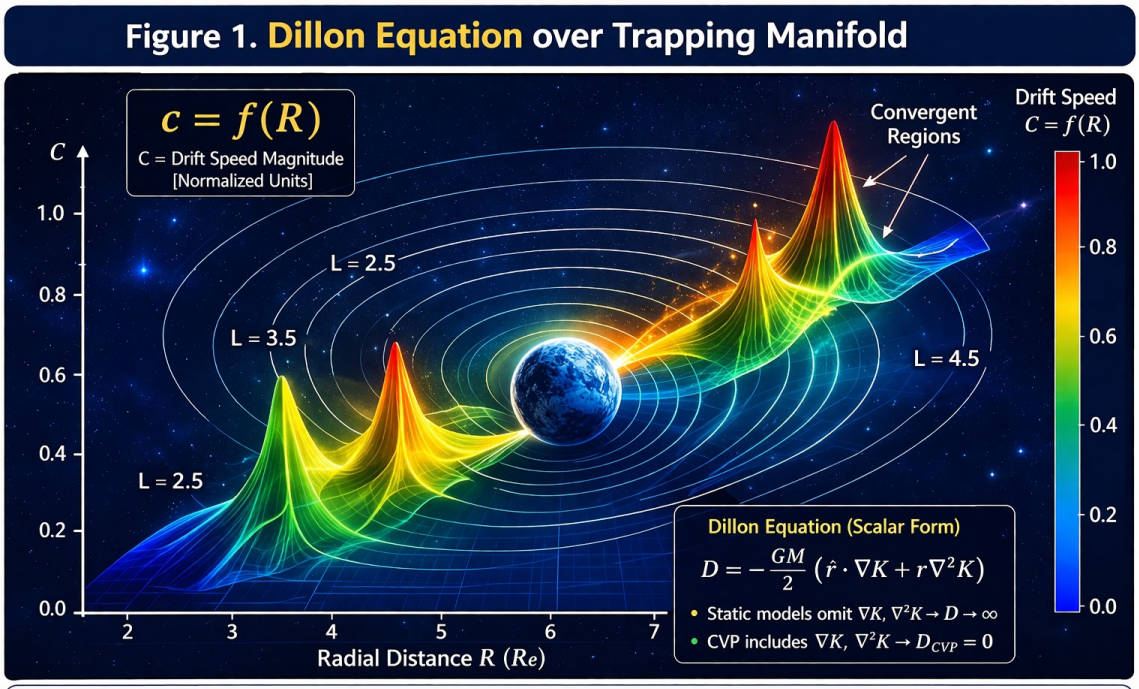

Figure 1: Cross-section of drift speed $C = f(R)$ over geodipolar L-shells.
Convergent regions surround "lumpy" drift-speed spikes in storm-driven turbulence.

4. Divergence and Instability in the Radiation Belts

To express geometric instability, this manuscript defines a magnetic divergence operator:

D_B = -(1/2)(L_hat · grad K_B + L Laplacian K_B).

This operator is the electromagnetic analogue of the divergence term used in the lunar CVP framework. It is intended to represent the instability introduced when the dominant curvature-gradient structure is omitted or under-resolved. Under disturbed conditions, sharp anisotropy in the magnetospheric field may cause D_B to increase rapidly, corresponding physically to storm-time flux blow-up, drift-shell splitting, rapid loss-cone emptying, or localized phase-space density spikes.

This interpretation is intentionally parallel to the gravitational CVP argument. In both domains, the failure mode is framed not as mere numerical inconvenience but as structural incompleteness of a model that underrepresents dynamic curvature.

5. Magnetic Response Operator and Proposed Convergence Condition

The corrective layer of the magnetospheric extension is the magnetic response operator:

C_B(K_B) = beta1 dK_B/dt + beta2 grad K_B + beta3 Laplacian K_B.

The temporal term captures dynamic response to storm-time compression and recovery, beta2 weights first-order spatial response such as radial transport and local manifold shear, and beta3 weights second-order smoothing of curvature discontinuities.

The formal convergence condition of the extension is div C_B(K_B) = Laplacian Phi_B(K_B), so that D_CVP,B = 0. Within the manuscript, this equation is the central compatibility statement. It is the magnetospheric analogue of the gravitational CVP closure condition and should be interpreted as a proposed

regularization law: if the response operator is assembled so that its divergence exactly balances the omitted-gradient instability, the corrected system is driven toward bounded evolution.

This does not by itself constitute empirical proof that the magnetosphere obeys the CVP-B formalism. What it does provide is a reviewable, mathematically explicit statement of how CVP would be tested in the radiation-belt setting.

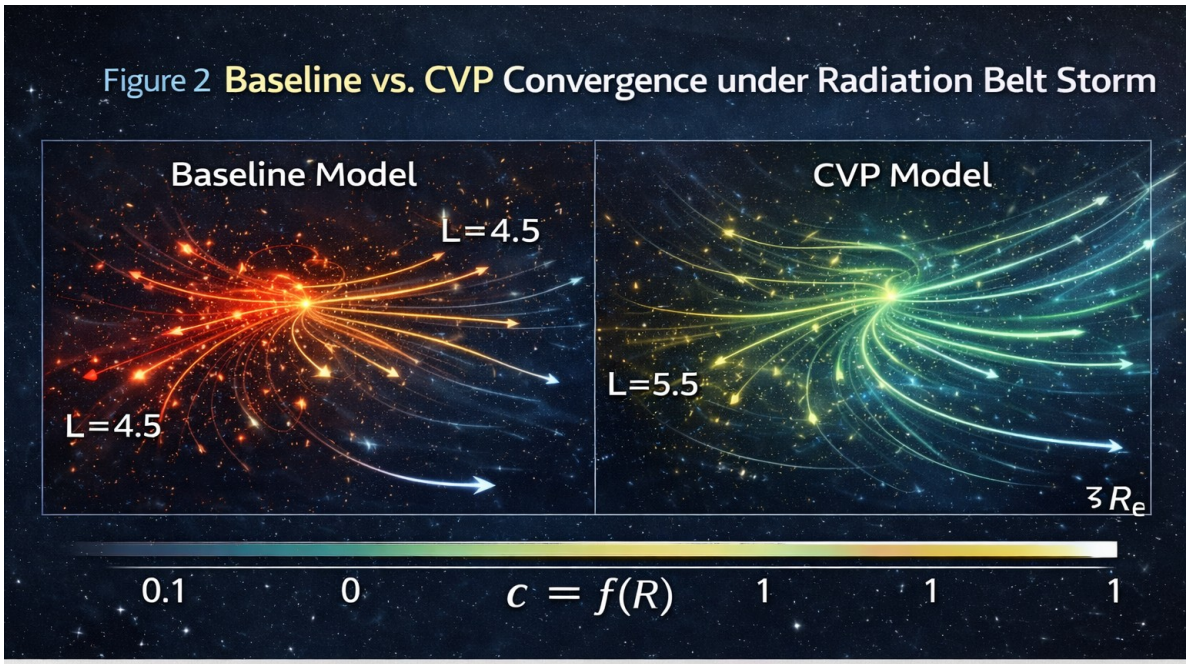

Figure 2: Phase-space flux comparison during dual storm peaks. Baseline model produces unbounded flux blow-up, while CVP enforces bounded and

6. Relation to Established Radiation-Belt Transport

A useful way to position CVP-B is as a curvature-variable closure layer on top of established transport formulations. A reduced phase-space description may be written as df/dt = -div J + S - L, where f is phase-space density, J is flux, and S and L are source and loss terms. In the CVP-B program, a curvature-associated flux divergence term may be isolated and paired with the response operator so that the corrected evolution becomes bounded under storm-time curvature distortion.

This positioning is deliberate. It avoids the overclaim that CVP-B supersedes quasilinear diffusion theory or the accumulated radiation-belt literature. Instead, it states a hypothesis: the trapping manifold admits a dynamic curvature representation whose response operator can serve as a mathematically compact regulator of otherwise unstable flux evolution.

7. Cross-Domain Mapping Between Gravitational and Magnetospheric CVP

Layer	Gravitational CVP	Magnetospheric CVP	Terminal geometry	Computational corridor
Curvature field	K(t, r, theta, phi)	K_B(t, L, alpha, phi)	-	-
Instability term	Divergence spike near irregular-body surface	Storm-time flux blow-up / curvature-manifold	-	-

		instability		
Response operator	C(K)	C_B(K_B)	-	-
Stable end-state	-	-	Stable landing site / post-storm bounded belt configuration	Altitude corridor / L-shell and pitch-angle corridor

Table 1. Structural mapping between the original gravitational CVP framework and the proposed magnetospheric extension.

8. Computational Implementation Path

The proposed extension is designed to be computationally tractable on the same tensor-based architecture already used for the gravitational CVP demonstrator. A GPU implementation would discretize an L-shell corridor, sample or infer K_B over the grid, evaluate grad K_B and Laplacian K_B numerically, assemble the operator C_B(K_B), and advance a trapped-particle state under corrected evolution.

The minimal runtime pattern is summarized in Table 2. This is not a full numerical methods section; it is a manuscript-level implementation note intended to show that the extension is executable rather than purely conceptual.

Runtime fact	Value
Implementation pattern	PyTorch tensor evaluation with optional dashboard layer
Field model	Dynamic magnetospheric curvature field K_B with storm-time perturbations
Core operators	grad K_B, Laplacian K_B, C_B(K_B), divergence status, attractor diagnostics
Execution target	CPU or CUDA GPU, depending on hardware availability
Working corridor	L-shell range 2.5-6.5; pitch-angle range 10°-80°

Table 2. Runtime facts for a proposed GPU-accelerated CVP-B pilot.

9. Figure Program for Review and Future Validation

A journal-ready manuscript should separate formalism from visualization while still making the intended validation path clear. The figure program mirrors the existing lunar manuscript structure. Figure 1 presents the magnetic-curvature topology over the L-shell and pitch-angle corridor, functioning as the electromagnetic analogue of the Dillon equation topology. Figure 2 provides a side-by-side quiver comparison of baseline magnetospheric divergence and CVP-B convergence, using an instability palette for the uncorrected field and a syntropy palette for the corrected field. Figure 3 presents a storm-time attractor diagram in which chaotic flux regimes collapse toward a bounded post-storm configuration.

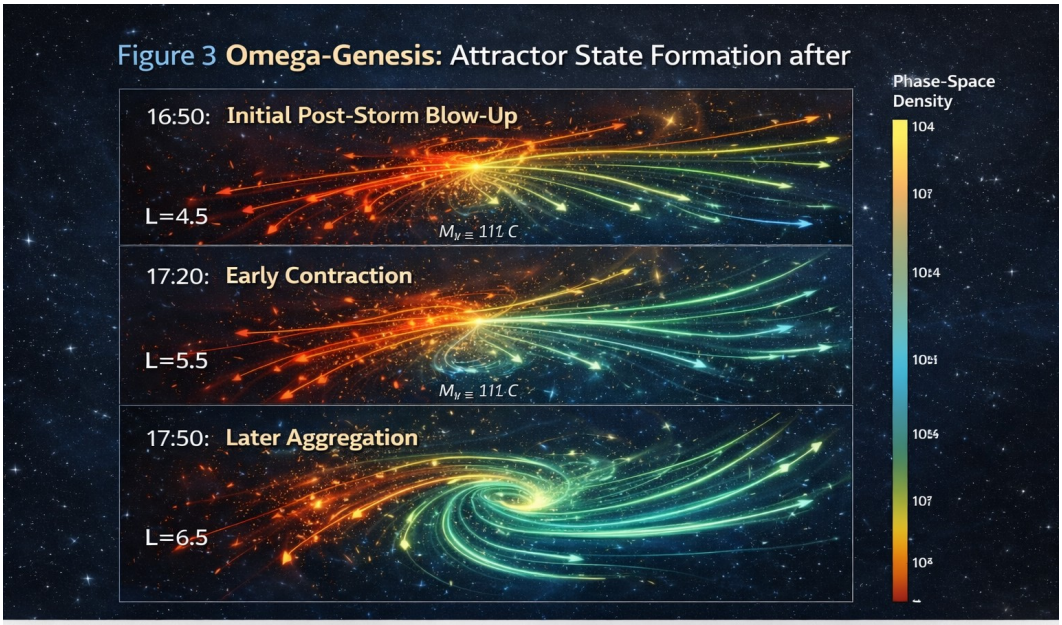

Figure 3: Phase-space density evolution shows storm-driven chaotic distributions condensing into a stable, *quasi-steady* attractor state *over time.*

10. Discussion

The strongest feature of the proposed extension is its operator continuity. The same mathematical grammar used to formalize instability and correction in irregular-body descent maps cleanly onto magnetospheric transport. That continuity makes the work attractive as a research program: the gravitational and electromagnetic domains can be investigated with parallel notation, shared computational infrastructure, and comparable divergence-versus-convergence diagnostics.

The principal limitation is equally clear. At present, CVP-B is a proposed formalism rather than an empirically validated radiation-belt theory. Any journal-ready presentation must therefore distinguish between established observational facts, such as storm-time belt variability, chorus-driven acceleration, and the documented behavior of the Van Allen belts, and the new curvature-variable closure proposed here. That distinction has been maintained throughout this manuscript.

If supported by simulation studies and comparison against Van Allen Probes, THEMIS, or related observational data sets, CVP-B could mature into a general-purpose curvature overlay for high-consequence magnetospheric modeling. If not, the present work would still have value as a mathematically disciplined comparison language between gravitational and electromagnetic stability problems.

11. Conclusion

This manuscript presents a standalone magnetospheric extension of Curvature Variable Physics for application to the Van Allen radiation belts. The framework is expressed as a clean research proposition: magnetic curvature is promoted to a dynamic field K_B(t, L, alpha, phi); omitted-gradient instability is represented by D_B; and a response operator C_B(K_B) is introduced to impose a formal convergence condition D_CVP,B = 0.

The paper retains only the formal narrative: modeling context, mathematical extension, computational path, cross-domain mapping, visualization program, and scope limitations. The next scientific step is straightforward: compare the CVP-B formalism against measured storm-time belt behavior using a GPU-accelerated implementation over the L-shell corridor.

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